

Min-Sum Diffusion vs. Dynamic Programming

Min-Sum Diffusion:

- converges to arc-consistency
- weakly convergent iterative algorithm
- + applicable to arbitrary graphs

Dynamic programming:

- finds arc-consistent (=integer) labeling
- + finite step algorithm (linear complexity)
- limited to acyclic graphs

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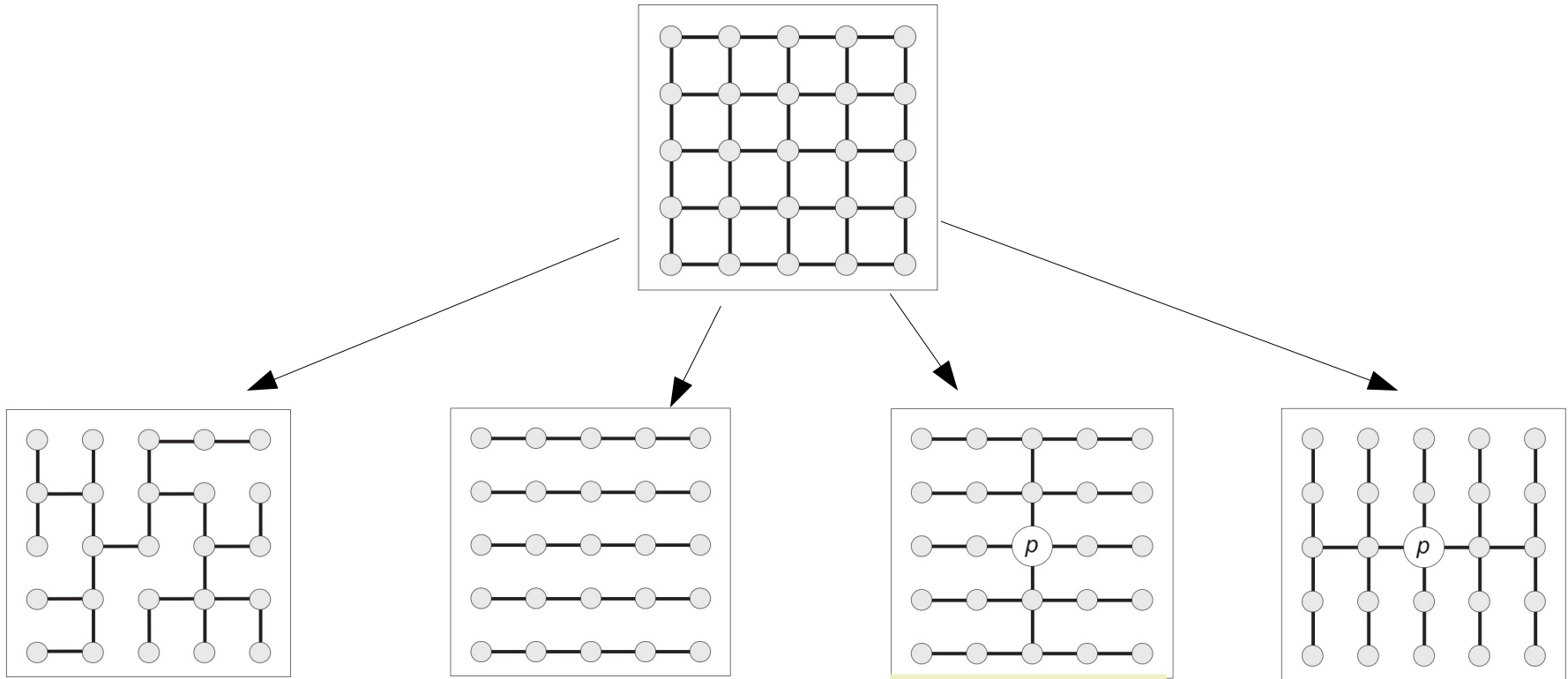
Dynamic programming:

- finds arc-consistent (=integer) labeling
- + finite step algorithm (linear complexity)
- limited to acyclic graphs

Our goal is a method, which:

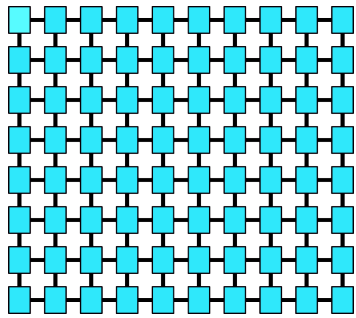
- addresses the dual LP and converges to arc-consistency
- + requires a finite number of steps for acyclic models
- + applicable to models with arbitrary graph structure

Acyclic Subgraphs



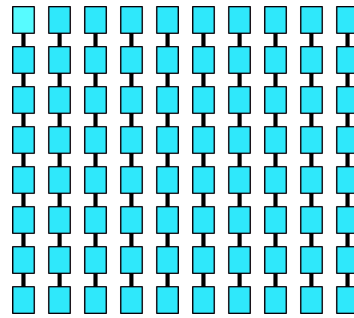
We can compute an optimal solution for any acyclic subgraph.
But what if they are inconsistent? How to combine them?

Lagrangian (Dual) Decomposition



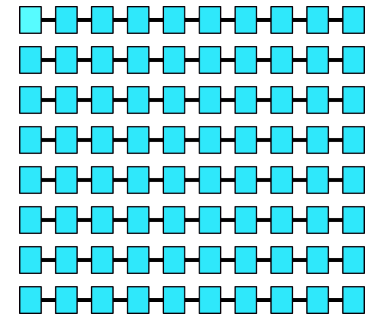
$$\langle \theta, \delta(x) \rangle$$

=



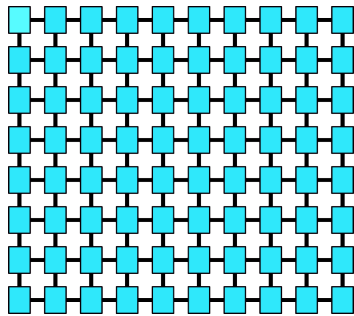
$$\langle \theta^c, \delta(x) \rangle$$

+

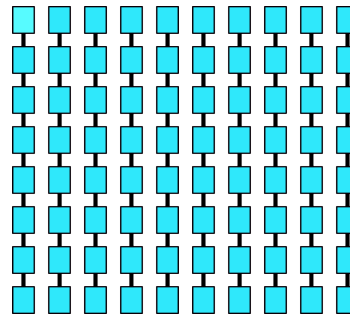


$$\langle \theta^r, \delta(x) \rangle$$

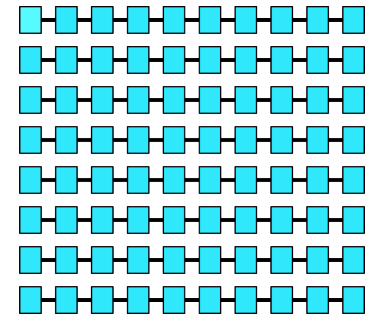
Lagrangian (Dual) Decomposition



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+



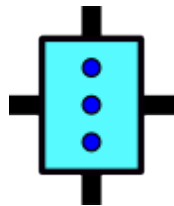
$$\langle \theta, \delta(x) \rangle$$

=

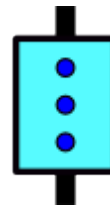
$$\langle \theta^c, \delta(x) \rangle$$

+

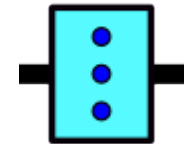
$$\langle \theta^r, \delta(x) \rangle$$



=



+



$$\theta_v(x_v)$$

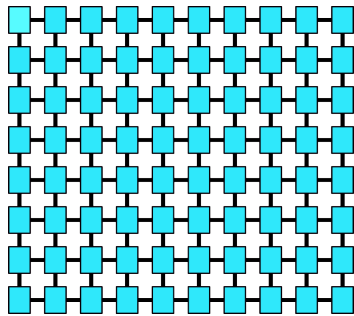
=

$$\frac{\theta_v(x_v)}{2} + \lambda_v(x_v)$$

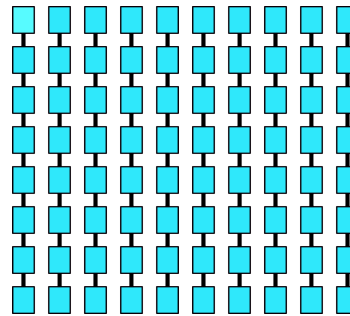
+

$$\frac{\theta_v(x_v)}{2} - \lambda_v(x_v)$$

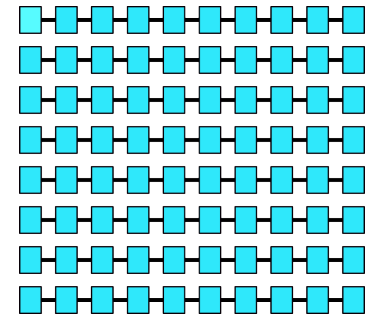
Lagrangian (Dual) Decomposition



=



+



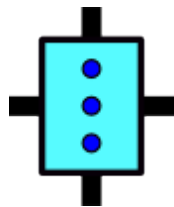
$$\langle \theta, \delta(x) \rangle$$

=

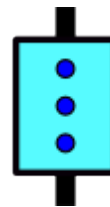
$$\langle \theta^c, \delta(x) \rangle$$

+

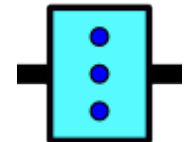
$$\langle \theta^r, \delta(x) \rangle$$



=



+



$$\theta_v(x_v)$$

=

$$\frac{\theta_v(x_v)}{2} + \lambda_v(x_v)$$

+

$$\frac{\theta_v(x_v)}{2} - \lambda_v(x_v)$$

$$\min_{x \in \mathcal{X}} \langle \theta, \delta(x) \rangle$$

$\forall \lambda$
 $\geq$

$$\min_{x^c \in \mathcal{X}^c} \langle \theta_{[\lambda]}^c, \delta(x^c) \rangle$$

+

$$\min_{x^r \in \mathcal{X}^r} \langle \theta_{[\lambda]}^r, \delta(x^r) \rangle$$

$$\begin{aligned} U(\lambda) &= \left[\min_{x^c \in \mathcal{X}^c} \left\langle \theta_{[\lambda]}^c, \delta(x^c) \right\rangle + \min_{x^r \in \mathcal{X}^r} \left\langle \theta_{[\lambda]}^r, \delta(x^r) \right\rangle \right] \\ &= \left[\min_{\mu^c \in \mathcal{L}^c} \left\langle \theta_{[\lambda]}^c, \mu^c \right\rangle + \min_{\mu^r \in \mathcal{L}^r} \left\langle \theta_{[\lambda]}^r, \mu^r \right\rangle \right] \end{aligned}$$

$$\theta_{[\lambda],v}^c = \frac{\theta_v}{2} + \lambda_v \quad \theta_{[\lambda],v}^r = \frac{\theta_v}{2} - \lambda_v \quad \theta_{[\lambda],uv}^c = \theta_{uv} \quad \theta_{[\lambda],uv}^r = \theta_{uv}$$

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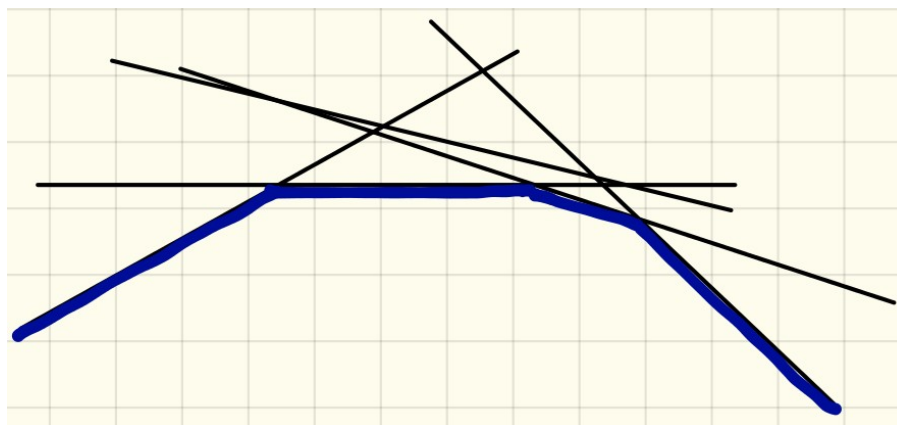
What are the properties of U ? :

- 1) Convex, differentiable
- 2) Convex, non-differentiable
- 3) Concave, piece-wise linear, non-differentiable
- 4) Concave, differentiable
- 5) None of the above
- 6) I do not know

Lagrangian (Dual) Decomposition

$$\begin{aligned} U(\lambda) &= \left[\min_{x^c \in \mathcal{X}^c} \left\langle \theta_{[\lambda]}^c, \delta(x^c) \right\rangle + \min_{x^r \in \mathcal{X}^r} \left\langle \theta_{[\lambda]}^r, \delta(x^r) \right\rangle \right] \\ &= \left[\min_{\mu^c \in \mathcal{L}^c} \left\langle \theta_{[\lambda]}^c, \mu^c \right\rangle + \min_{\mu^r \in \mathcal{L}^r} \left\langle \theta_{[\lambda]}^r, \mu^r \right\rangle \right] \end{aligned}$$

$$\theta_{[\lambda],v}^c = \frac{\theta_v}{2} + \lambda_v \quad \theta_{[\lambda],v}^r = \frac{\theta_v}{2} - \lambda_v \quad \theta_{[\lambda],uv}^c = \theta_{uv} \quad \theta_{[\lambda],uv}^r = \theta_{uv}$$



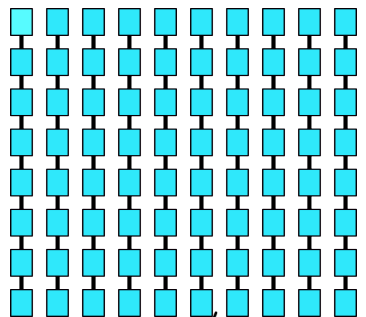
U is piece-wise linear, concave

Lagrangian (Dual) Decomposition

$$U(\lambda) = \left[\min_{\mu^c \in \mathcal{L}^c} \left\langle \theta_{[\lambda]}^c, \mu^c \right\rangle + \min_{\mu^r \in \mathcal{L}^r} \left\langle \theta_{[\lambda]}^r, \mu^r \right\rangle \right]$$

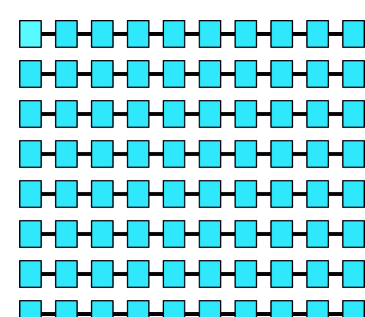
$$\theta_{[\lambda],v}^c = \frac{\theta_v}{2} + \lambda_v \quad \theta_{[\lambda],v}^r = \frac{\theta_v}{2} - \lambda_v \quad \theta_{[\lambda],uv}^c = \theta_{uv} \quad \theta_{[\lambda],uv}^r = \theta_{uv}$$

$$\frac{\partial U}{\partial \lambda_v} = \mu_v^{c*} - \mu_v^{r*}$$



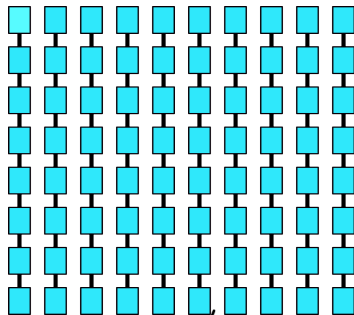
$$\mu^{c*} = \min_{\mu^c \in \mathcal{L}^c} \left\langle \theta_{[\lambda]}^c, \mu^c \right\rangle$$

← Slave problems/solutions →

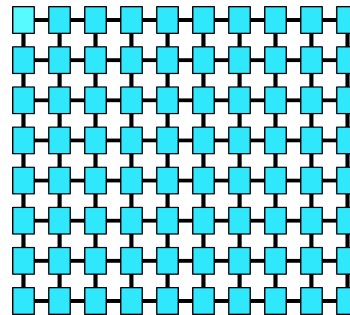


$$\mu^{r*} = \min_{\mu^r \in \mathcal{L}^r} \left\langle \theta_{[\lambda]}^r, \mu^r \right\rangle$$

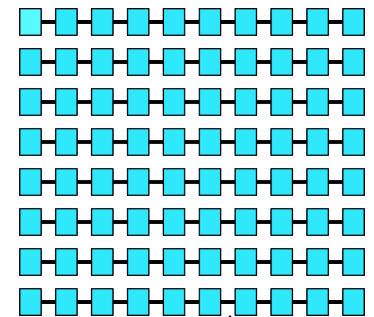
$$\left(\frac{\partial U}{\partial \lambda_v} = \mu_v^{c*} - \mu_v^{r*} = 0 \right) \Rightarrow \left(\mu_v^{c*} = \mu_v^{r*} \right)$$



$$\mu^{c*} = \min_{\mu^c \in \mathcal{L}^c} \left\langle \theta_{[\lambda]}^c, \mu^c \right\rangle$$



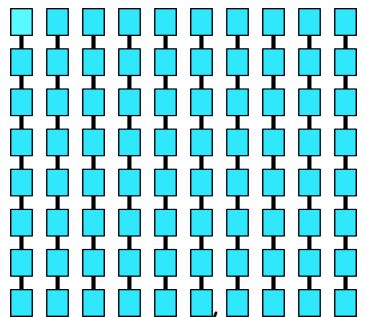
$$\mu^* = (\mu_v^c, \mu_{uv}^c, \mu_{uv}^r)$$

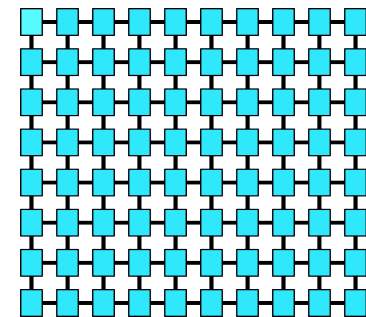


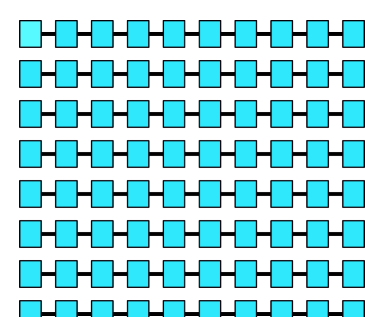
$$\mu^{r*} = \min_{\mu^r \in \mathcal{L}^r} \left\langle \theta_{[\lambda]}^r, \mu^r \right\rangle$$

Proposition: $\mu^* = \arg \min_{\mu \in \mathcal{L}} \langle \theta, \mu \rangle$

$$\left(\frac{\partial U}{\partial \lambda_v} = \mu_v^{c*} - \mu_v^{r*} = 0 \right) \Rightarrow \left(\mu_v^{c*} = \mu_v^{r*} \right)$$


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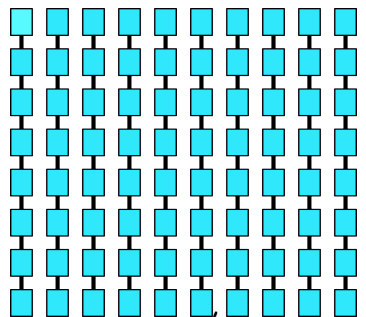

$$\mu^* = (\mu_v^c, \mu_{uv}^c, \mu_{uv}^r)$$


$$\mu^{r*} = \min_{\mu^r \in \mathcal{L}^r} \langle \theta_{[\lambda]}^r, \mu^r \rangle$$

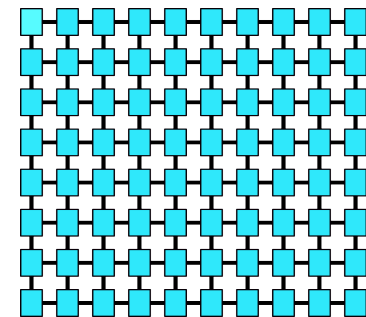
Proposition: $\mu^* = \arg \min_{\mu \in \mathcal{L}} \langle \theta, \mu \rangle$

Lemma: $\min_{\mu \in \mathcal{L}} \langle \theta, \mu \rangle \geq U(\lambda) = \left[\min_{\mu^c \in \mathcal{L}^c} \langle \theta_{[\lambda]}^c, \mu^c \rangle + \min_{\mu^r \in \mathcal{L}^r} \langle \theta_{[\lambda]}^r, \mu^r \rangle \right]$

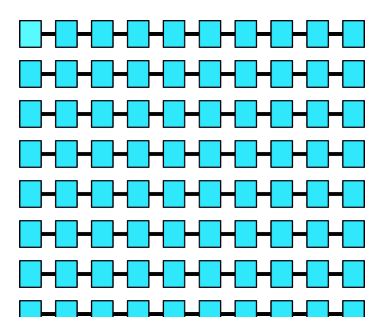
$$\left(\frac{\partial U}{\partial \lambda_v} = \mu_v^{c*} - \mu_v^{r*} = 0 \right) \Rightarrow \left(\mu_v^{c*} = \mu_v^{r*} \right)$$



$$\mu^{c*} = \min_{\mu^c \in \mathcal{L}^c} \langle \theta_{[\lambda]}^c, \mu^c \rangle$$



$$\mu^* = (\mu_v^c, \mu_{uv}^c, \mu_{uv}^r)$$

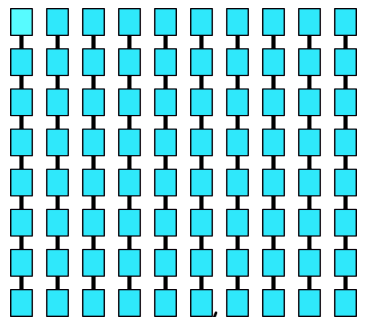


$$\mu^{r*} = \min_{\mu^r \in \mathcal{L}^r} \langle \theta_{[\lambda]}^r, \mu^r \rangle$$

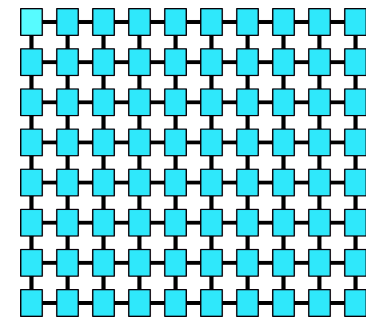
Proposition: $\mu^* = \arg \min_{\mu \in \mathcal{L}} \langle \theta, \mu \rangle$

Proof:
$$\min_{\mu \in \mathcal{L}} \langle \theta, \mu \rangle \geq U(\lambda) = \left[\min_{\mu^c \in \mathcal{L}^c} \langle \theta_{[\lambda]}^c, \mu^c \rangle + \min_{\mu^r \in \mathcal{L}^r} \langle \theta_{[\lambda]}^r, \mu^r \rangle \right]$$
$$= \langle \theta_{[\lambda]}^c, \mu^{c*} \rangle + \langle \theta_{[\lambda]}^r, \mu^{r*} \rangle = \langle \theta, \mu^* \rangle$$

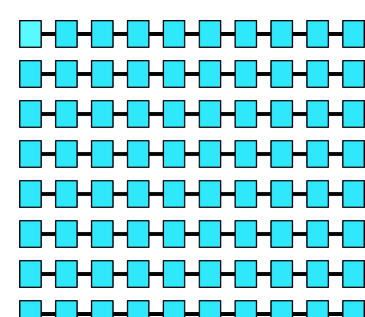
$$\left(\frac{\partial U}{\partial \lambda_v} = \mu_v^{c*} - \mu_v^{r*} = 0 \right) \Rightarrow \left(\mu_v^{c*} = \mu_v^{r*} \right)$$



$$\mu^{c*} = \min_{\mu^c \in \mathcal{L}^c} \langle \theta_{[\lambda]}^c, \mu^c \rangle$$



$$\mu^* = (\mu_v^c, \mu_{uv}^c, \mu_{uv}^r)$$



$$\mu^{r*} = \min_{\mu^r \in \mathcal{L}^r} \langle \theta_{[\lambda]}^r, \mu^r \rangle$$

Proposition: $\mu^* = \arg \min_{\mu \in \mathcal{L}} \langle \theta, \mu \rangle$

Proof:
$$\min_{\mu \in \mathcal{L}} \langle \theta, \mu \rangle \geq U(\lambda) = \left[\min_{\mu^c \in \mathcal{L}^c} \langle \theta_{[\lambda]}^c, \mu^c \rangle + \min_{\mu^r \in \mathcal{L}^r} \langle \theta_{[\lambda]}^r, \mu^r \rangle \right]$$
$$= \langle \theta_{[\lambda]}^c, \mu^{c*} \rangle + \langle \theta_{[\lambda]}^r, \mu^{r*} \rangle = \langle \theta, \mu^* \rangle$$

$$\left(\frac{\partial U}{\partial \lambda_v} = \mu_v^{c*} - \mu_v^{r*} = 0 \right) \Rightarrow \left(\mu_v^{c*} = \mu_v^{r*} \right)$$

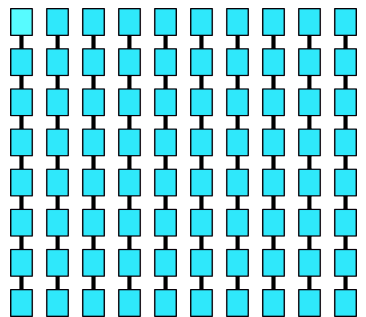
Maximum of the decomposition-based dual = Minimum of the relaxed primal !

Lagrangian (Dual) Decomposition

$$U(\lambda) = \left[\min_{\mu^c \in \mathcal{L}^c} \left\langle \theta_{[\lambda]}^c, \mu^c \right\rangle + \min_{\mu^r \in \mathcal{L}^r} \left\langle \theta_{[\lambda]}^r, \mu^r \right\rangle \right]$$

$$\theta_{[\lambda],v}^c = \frac{\theta_v}{2} + \lambda_v \quad \theta_{[\lambda],v}^r = \frac{\theta_v}{2} - \lambda_v \quad \theta_{[\lambda],uv}^c = \theta_{uv} \quad \theta_{[\lambda],uv}^r = \theta_{uv}$$

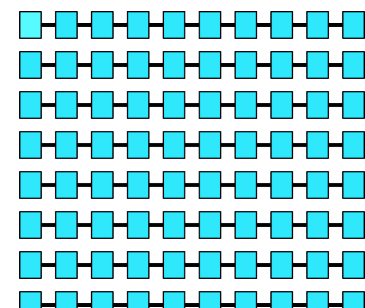
$$\frac{\partial U}{\partial \lambda_v} = \mu_v^{c*} - \mu_v^{r*} \quad \text{in particular} \quad \frac{\partial U}{\partial \lambda_v} = \delta(x^{c*})_v - \delta(x^{r*})_v$$



$$x^{c*} = \min_{x \in \mathcal{X}^c} \left\langle \theta_{[\lambda]}^c, \delta(x) \right\rangle$$

Column-wise labeling

← Slave problems/solutions →



$$x^{r*} = \min_{x \in \mathcal{X}^r} \left\langle \theta_{[\lambda]}^r, \delta(x) \right\rangle$$

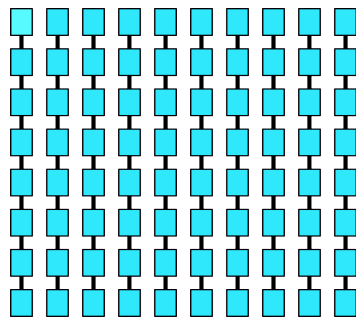
Row-wise labeling

Optimality Conditions: Non-Relaxed Optimality

$$U(\lambda) = \left[\min_{\mu^c \in \mathcal{L}^c} \langle \theta_{[\lambda]}^c, \mu^c \rangle + \min_{\mu^r \in \mathcal{L}^r} \langle \theta_{[\lambda]}^r, \mu^r \rangle \right] \quad \frac{\partial U}{\partial \lambda_v} = \delta(x^{c*})_v - \delta(x^{r*})_v$$

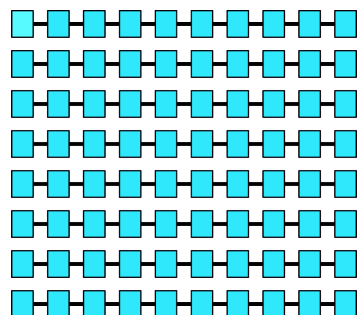
Proposition:

$$\left(\exists x^{c*}, x^{r*} : x^{c*} = x^{r*} = x^* \right) \Rightarrow \left(x^* = \arg \min_{x \in \mathcal{X}_v} \langle \theta, \delta(x) \rangle \right)$$


$$x^{c*} = \min_{x \in \mathcal{X}^c} \langle \theta_{[\lambda]}^c, \delta(x) \rangle$$

Column-wise labeling

← Slave problems/solutions →


$$x^{r*} = \min_{x \in \mathcal{X}^r} \langle \theta_{[\lambda]}^r, \delta(x) \rangle$$

Row-wise labeling

Optimality Conditions: Non-Relaxed Optimality

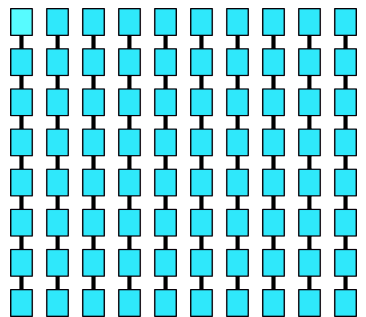
$$U(\lambda) = \left[\min_{\mu^c \in \mathcal{L}^c} \langle \theta_{[\lambda]}^c, \mu^c \rangle + \min_{\mu^r \in \mathcal{L}^r} \langle \theta_{[\lambda]}^r, \mu^r \rangle \right] \quad \frac{\partial U}{\partial \lambda_v} = \delta(x^{c*})_v - \delta(x^{r*})_v$$

Proposition:

$$\left(\exists x^{c*}, x^{r*} : x^{c*} = x^{r*} = x^* \right) \Rightarrow \left(x^* = \arg \min_{x \in \mathcal{X}_v} \langle \theta, \delta(x) \rangle \right)$$

Proof: $\min_{x \in \mathcal{X}} \langle \theta, \delta(x) \rangle \geq \min_{\mu^c \in \mathcal{L}^c} \langle \theta_{[\lambda]}^c, \mu^c \rangle + \min_{\mu^r \in \mathcal{L}^r} \langle \theta_{[\lambda]}^r, \mu^r \rangle$

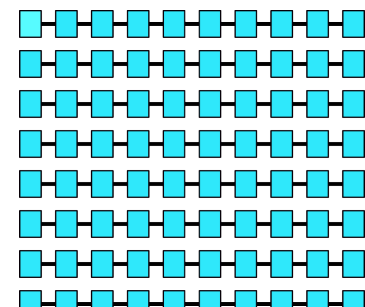
$$= \langle \theta_{[\lambda]}^c, \delta(x^*) \rangle + \langle \theta_{[\lambda]}^r, \delta(x^*) \rangle = \langle \theta, \delta(x^*) \rangle$$



$$x^{c*} = \min_{x \in \mathcal{X}^c} \langle \theta_{[\lambda]}^c, \delta(x) \rangle$$

Column-wise labeling

← Slave problems/solutions →



$$x^{r*} = \min_{x \in \mathcal{X}^r} \langle \theta_{[\lambda]}^r, \delta(x) \rangle$$

Row-wise labeling

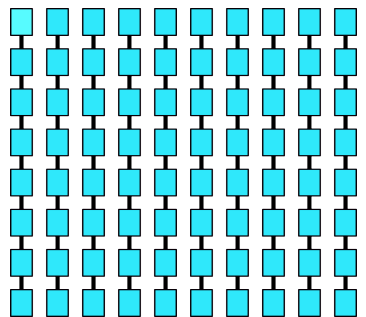
Optimality Conditions: Non-Relaxed Optimality

$$U(\lambda) = \left[\min_{\mu^c \in \mathcal{L}^c} \langle \theta_{[\lambda]}^c, \mu^c \rangle + \min_{\mu^r \in \mathcal{L}^r} \langle \theta_{[\lambda]}^r, \mu^r \rangle \right] \quad \frac{\partial U}{\partial \lambda_v} = \delta(x^{c*})_v - \delta(x^{r*})_v$$

Proposition:

$$\left(\exists x^{c*}, x^{r*} : x^{c*} = x^{r*} = x^* \right) \Rightarrow \left(x^* = \arg \min_{x \in \mathcal{X}_v} \langle \theta, \delta(x) \rangle \right)$$

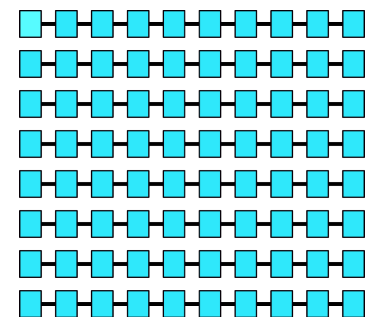
Equal slave integer labelings = Optimal integer labeling in the master problem



$$x^{c*} = \min_{x \in \mathcal{X}^c} \langle \theta_{[\lambda]}^c, \delta(x) \rangle$$

Column-wise labeling

← Slave problems/solutions →



$$x^{r*} = \min_{x \in \mathcal{X}^r} \langle \theta_{[\lambda]}^r, \delta(x) \rangle$$

Row-wise labeling